CHEMISTRY ADAPTLY

ORGANIC CHEMISTRY – FRACTIONAL DISTILLATION

KEY VOCABULARY

Fraction: a group of hydrocarbons with similar boiling points.
Fractionating column: hot at the bottom, cool at the top.
Condense: a vapour turns to liquid at its boiling point.
Volatility: how easily a substance evaporates.

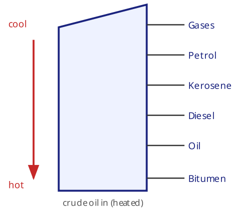
COMMON MISCONCEPTIONS

~~Fractional distillation is a chemical reaction.~~
✓ It is a physical separation — no new substances.

~~Small molecules collect at the bottom.~~
✓ Small, low-bp molecules rise to the cool top.

~~The column is the same temperature throughout.~~
✓ It has a temperature gradient: hot below, cool above.

FRACTIONATING COLUMN



1 WHY IT WORKS

Explain how fractional distillation separates crude oil. **[3 marks]**

Refer to boiling points and the temperature gradient.

2 TOP OR BOTTOM

State where the **shortest molecules** leave the column and why. **[2 marks]**

3 PROPERTIES

Explain why bitumen is more viscous than petrol. **[2 marks]**

4 NAME FRACTIONS

Name two useful fractions obtained from crude oil. **[2 marks]**

5 FILL THE GAPS

Complete the sentence using the word bank. **[2 marks]**

Each fraction _____ at the height where the temperature equals its _____ point.

boiling | condense | melting | freeze

6 PHYSICAL CHANGE

Explain why fractional distillation is a physical process. **[1 mark]**

TIP: *Hot bottom, cool top: short molecules rise high, long molecules stay low.*

✓ **Think: Why do fractions contain a range of molecule sizes rather than just one?**